

Semester: **Fall 2022**
Course Code: **CSE460**
Course Title: **VLSI Design**

Midterm Exam
Full Marks: **15 x 3 = 45**
Time: **1 hour 30 minutes**
Date: **11th November 2022**

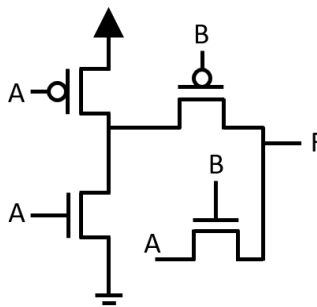
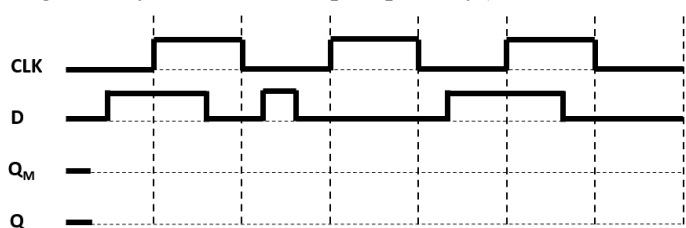
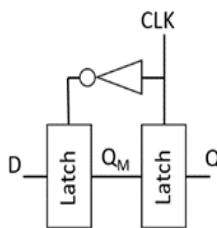
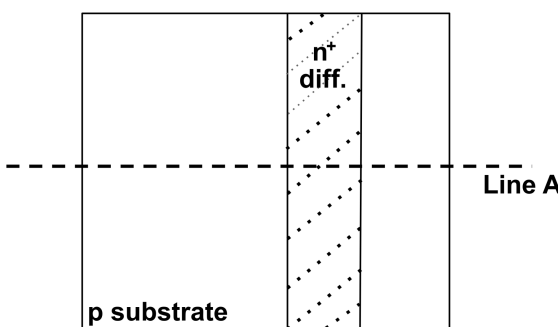
Set B

Student ID:	Name:	Section:
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[Answer any **THREE** questions out of **FOUR**. Each question carries equal marks.]

[After the exam, the question paper should be turned in along with the answer script.]

1. (CO2)	(a)	Design a CMOS compound gate that implements the following function: $Y = (A.B + C.D) . E$	6
	(b)	Imagine a <i>gate-level</i> logic circuit with 1 output f , and 4 inputs x1 , x0 , y1 , and y0 . - Let $X = x1 x0$ be a two-digit binary number, where the four possible values of X are 00, 01, 10, and 11, which represent the four decimal numbers 0, 1, 2, and 3 respectively. - Let $Y = y1 y0$ be another number with the same four possible values. - The output f should be 1 if the sum of numbers represented by X and Y is <i>less</i> than decimal 3. Otherwise, the output f should be 0.	
		i) Create a truth table for all 16 combinations of the 4 inputs.	3
		ii) Derive the boolean expression of the output using k-map.	4
		iii) Implement the function f with logic gates.	2
2. (CO2)	Given is a pattern for a sequence detector.		
	CLK	t₁ t₂ t₃ t₄ t₅ t₆ t₇ t₈ t₉ t₁₀ t₁₁ t₁₂ t₁₃ t₁₄ t₁₅ t₁₆ t₁₇ t₁₈	
	w	1 1 1 0 1 0 1 1 0 0 1 0 1 0 0 0 1 0	
	z	0 1 1 0 0 0 0 1 0 1 0 0 0 0 1 1 0 0	
	(a) Identify the <u>sequence</u> and the <u>type</u> of FSM for the above input-output combinations. [Hint: The above table contains two sequences]		1+1
	(b) Design the state diagram, state table, <i>gray encoded</i> state assigned table. [Overlapping allowed].		3+2+2
	(c) Determine the logic expressions of the next state and output variables using K-map.		6
3. (CO2)	(a)	Complete the truth table shown in Figure 2 for the circuit shown in Figure 1	5

		<table border="1" data-bbox="876 241 1131 524"><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td></td></tr><tr><td>0</td><td>1</td><td></td></tr><tr><td>1</td><td>0</td><td></td></tr><tr><td>1</td><td>1</td><td></td></tr></table>	A	B	F	0	0		0	1		1	0		1	1		
A	B	F																
0	0																	
0	1																	
1	0																	
1	1																	
	Figure 1: A Logic Circuit	Figure 2: Truth Table																
(b)	A D flip-flop is made by cascading two D latches as shown in Figure 3. Data input (D) and clock (CLK) for the circuit are shown in Figure 4. Draw the waveforms for Q_M and Q. (Use a pencil to draw the signals and draw on the question paper, no need to draw the diagram in your answer script separately.) 	9																
(c)	Identify the type of the flip-flop shown in Figure 3.	1																
4. (CO1)	Observe figure 5 and answer the following questions.																	
																		
	Figure 5: Top view of a wafer after fabricating an n+ diffusion region (dotted region)																	
(a)	Draw the cross-sectional view of the above figure along ‘Line A’ and label the different regions.	3																
(b)	Briefly describe and illustrate the necessary steps to fabricate the n+ diffusion region on a blank wafer as shown in the figure.	12																